OOP Practical No: 06

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INPUT:

```
#include <iostream>
using namespace std;

class complex
{
   float real, img;

public:
   void accept(float r, float i)
   {
      real = r;
      img = i;
   }

   void display()
   {
      if (img >= 0)
         cout << real << "+i" << img;
      else
         cout << real << "-i" << -img;
   }

   complex operator+(complex b)
   {
      complex sum;
```

```
    sum.real = real + b.real;
    sum.img = img + b.img;
    return sum;
}

complex operator-(complex b)
{
    complex sub;
    sub.real = real - b.real;
    sub.img = img - b.img;
    return sub;
}

complex operator*(complex b)
{
    complex mul;
    mul.real = (real * b.real) - (img * b.img);
    mul.img = (real * b.img) + (img * b.real);
    return mul;
}

complex operator/(complex b)
{
    complex div;

    float d = b.real * b.real + b.img * b.img;

    if (d == 0)
    {
        cout << "\nDivision by zero is not possible.";
        div.real = 0;
```

```
            div.img = 0;
            return div;
        }

        div.real = (real * b.real + img * b.img) / d;
        div.img = (img * b.real - real * b.img) / d;

        return div;
    }
};

int main()
{
    complex c1, c2, c3, c4, c5, c6;

    c1.accept(2, 4);
    c2.accept(6, 3);

    cout << "c1 = ";
    c1.display();

    cout << "\nc2 = ";
    c2.display();

    c3 = c1 + c2;
    cout << "\n\nThe addition of 2 complex numbers is:\n";
    c3.display();

    c4 = c1 - c2;
    cout << "\n\nThe subtraction of 2 complex numbers is:\n";
    c4.display();
```

```
    c5 = c1 * c2;
    cout << "\n\nThe multiplication of 2 complex numbers is:\n";
    c5.display();

    c6 = c1 / c2;
    cout << "\n\nThe division of 2 complex numbers is:\n";
    c6.display();

    cout << endl;

    return 0;
}
```

OUTPUT:

```
PS D:\ENGINEERING\SEM I\OOP\Practical-No-06-Operator-overloading> .\code.exe
c1 = 2+i4
c2 = 6+i3

The addition of 2 complex numbers is:
8+i7

The subtraction of 2 complex numbers is:
-4+i1

The multiplication of 2 complex numbers is:
0+i30

The division of 2 complex numbers is:
0.533333+i0.4
PS D:\ENGINEERING\SEM I\OOP\Practical-No-06-Operator-overloading>
```